

# STADIOchoice — Data Request

## Capstone Project (CAP182) — Subscriber Churn Prediction

To predict subscriber churn for STADIOstream, the following data is requested from STADIOchoice. The request is split into three related tables, which can be linked using a common Subscriber ID: **Subscription & Billing Data**, **Streaming Viewing Behaviour Data**, and **Cancellation & Retention Data**.

**Table 1: Subscription & Billing Data**

| Column name                                | Data type           | Example                               | Additional information                                                                             |
|--------------------------------------------|---------------------|---------------------------------------|----------------------------------------------------------------------------------------------------|
| Subscriber ID                              | String, categorical | SUB00123456                           | A unique ID per subscriber. Must not include the subscriber's real name or ID number, for privacy. |
| Plan type                                  | String, categorical | "Premium", "Standard", "Basic"        | List all valid plan names STADIOstream currently offers.                                           |
| Monthly price                              | Numeric, float      | 199.00                                | In South African Rand (ZAR), the amount billed per month.                                          |
| Subscription start date                    | Date                | 2023-04-15                            | Format: yyyy-MM-dd. The date the subscriber first joined STADIOstream.                             |
| Tenure (months)                            | Numeric, integer    | 14                                    | Number of months the subscriber has been active, calculated up to the extract date.                |
| Payment status                             | String, categorical | "Success", "Failed", "Pending"        | Status of the most recent billing attempt.                                                         |
| Number of payment failures (last 6 months) | Numeric, integer    | 2                                     | Used to detect subscribers at risk due to payment issues.                                          |
| Upgrade/downgrade history                  | String, categorical | "Upgraded", "Downgraded", "No change" | Whether the subscriber changed their plan in the last 6 months.                                    |
| Free trial flag                            | Boolean             | TRUE / FALSE                          | Whether the subscriber is currently on a free trial rather than a paid plan.                       |

**Table 2: Streaming Viewing Behaviour Data**

| Column name                              | Data type           | Example     | Additional information                                                                         |
|------------------------------------------|---------------------|-------------|------------------------------------------------------------------------------------------------|
| Subscriber ID                            | String, categorical | SUB00123456 | Must match the Subscriber ID in Table 1, so the two tables can be linked together.             |
| Titles watched (last 30 days)            | Numeric, integer    | 6           | Count of distinct titles the subscriber watched in the last 30 days.                           |
| Total watch time (last 30 days, minutes) | Numeric, float      | 420.5       | Total minutes streamed in the last 30 days.                                                    |
| Days since last login                    | Numeric, integer    | 9           | Number of days since the subscriber last opened the app. A high number suggests disengagement. |

| Column name                 | Data type           | Example                               | Additional information                                                                                |
|-----------------------------|---------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------|
| Search count (last 30 days) | Numeric, integer    | 3                                     | Number of searches performed, including searches with zero results.                                   |
| Zero-result search flag     | Boolean             | TRUE / FALSE                          | Whether any of the subscriber's recent searches returned no matching content (a sign of frustration). |
| Device type                 | String, categorical | "Mobile", "Smart TV", "Web", "Tablet" | The main device used to stream, if available.                                                         |

**Table 3: Cancellation & Retention Data**

| Column name                    | Data type           | Example                                                                         | Additional information                                                                                      |
|--------------------------------|---------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Subscriber ID                  | String, categorical | SUB00123456                                                                     | Must match Subscriber ID in Tables 1 and 2.                                                                 |
| Churned (target variable)      | Boolean             | TRUE / FALSE                                                                    | TRUE if the subscriber cancelled during the observation period; this is the outcome the model will predict. |
| Cancellation date              | Date                | 2024-02-10                                                                      | Format: yyyy-MM-dd. Leave blank if the subscriber has not cancelled.                                        |
| Cancellation reason (if given) | String, categorical | "Too expensive", "Not enough content", "Technical issues", "Other", "Not given" | Free-text reasons should be grouped into categories where possible.                                         |
| Retention offer made           | Boolean             | TRUE / FALSE                                                                    | Whether STADIOchoice offered a discount or free month to try to keep the subscriber.                        |
| Retention offer accepted       | Boolean             | TRUE / FALSE                                                                    | Whether the subscriber accepted the retention offer, if one was made.                                       |